Human-in-the-Loop Grading for Mathematics Homework

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Abstract

We study how large language models can produce mathematical arguments whose intermediate claims are individually checkable and traceable to evidence. Our benchmark combines symbolic verification, retrieval over source documents, and expert review. Results suggest that explicit evidence anchors reduce unsupported claims without sacrificing coverage.

1. Introduction

Mathematical reasoning is often evaluated only by its final answer. This paper argues that the intermediate chain must also be auditable. We define a verifiable claim as a statement with a local justification, a declared set of assumptions, and an evidence reference when external knowledge is used.

2. Method

Given a problem P and a candidate proof s1,...,sn, each step is checked by a deterministic symbolic tool. If a step cannot be verified, it is marked as requiring additional conditions rather than accepted. The evidence layer links each external claim to a source span and a retrieval score.

3. Findings

Across 240 annotated solutions, evidence anchoring lowered the rate of unsupported intermediate claims from 31% to 12%. Human reviewers reported that page-level evidence chips made verification substantially faster. Remaining errors cluster around implicit domain assumptions.

4. Discussion

Verifiability is not the same as correctness. A proof can be formally checkable yet rely on an undeclared assumption. A useful research workspace should therefore keep assumptions, citations, and source locations attached to the same project object.